

Hermes F&O Skills & Strategy Library

Final research-integrated edition: gap strategies from the supplied 19-page source, current NSE opening mechanics, and additional quantitative methods supported by academic or exchange research.

How to use this manual: Treat every skill as a testable module. A historical article can inspire a hypothesis; it cannot be used as proof that the same rule works today or on NSE.

1. What the New Source Adds

- The uploaded source contains four gap concepts: Morning Reversal, Overnight Gap, Opening Gap and Gap Closer. [source]
- Its Morning Reversal test found substantial partial and full gap retracement in a historical 1999-2003 US index sample. The proper lesson for Hermes is to test the phenomenon, not transplant its old percentages or timestamps to India. [source]
- The same source contains both reversal and continuation logic: an opening gap can fade, but a gap can also remain open for days and become a later continuation setup. [source]
- Its Gap Closer experiment is especially valuable as a risk lesson: 50 of 62 gaps closed for profit, but the minority that stayed open generated large losses, leading the source itself to warn that gap-only trading entails significant risk. [source]

2. Gap Strategy Library for Hermes

Module	Hypothesis	Core features	Failure mode
GAP-FILL	Opening gap tends to retrace toward prior close/gap boundary	GapPct, GapATR, regime, OFI, VWAP, breadth	Strong news/trend gap continues
GAP-AND-GO	Gap acceptance leads to continuation	Gap, opening drive, relative volume, breadth, futures	False breakout / exhaustion
GAP-FADE	Gap rejects and reverses	Gap, failed auction, CVD/OFI, VWAP	Trend day
GAP-MIDPOINT	Half-gap is a useful probabilistic target	Gap size, distance, time	Reversal stalls before midpoint
GAP-CLOSER	Older gaps can become future target zones	Gap ledger, age, distance, ATR	Gap never closes / large drawdown
MULTI-DAY GAP	Unfilled historical gap interacts with current structure	Gap age, retests, profile, volume	Structural break
ISLAND REVERSAL	Opposite gap after an intervening range can mark reversal	Daily/intraday gaps + volume	News shock overrides setup
OPENING DRIVE	Acceptance beyond early range indicates direction	Opening range, delta, breadth, VWAP	Whipsaw at open

3. Gap Rules That Must Be Modernized for NSE

- The supplied gap articles use old US instruments, US tick sizes, US pre-market share-count thresholds and US Eastern-time session clocks. Those exact thresholds should not be copied into Hermes.
- NSE currently runs a 09:00-09:15 pre-open process and a 09:15-15:40 equity-derivatives regular session. [source]
- Hermes should replace fixed historical rules with normalized variables: GapPct, GapATR, pre-open buy/sell imbalance, imbalance persistence, relative pre-open volume, first-5/15/30-minute range, VWAP distance, OFI, breadth, futures basis/OI, option IV/skew/gamma context, and event flags.
- Historical US observations such as a reversal beginning around 9:36 ET or peaking around 9:53 ET are useful as examples of time-of-day modelling, not India trading instructions. [source]

4. Opening Auction Intelligence - High Priority

- NSE's pre-open is a call-auction process. During order collection, orders can be entered/modified/cancelled; matching determines the opening price. NSE also disseminates indicative equilibrium price and quantities.

[source]

- Build Opening Auction Imbalance: $OAI = (BuyQty - SellQty) / (BuyQty + SellQty)$. Store its level, slope, acceleration, persistence and distance from the eventual open.
- Model the probability that pre-open imbalance survives into the first 1, 5 and 15 minutes. This creates a clean bridge between auction information and gap-and-go/gap-fade decisions.
- Opening-auction research shows that the open is a major price-discovery event following the non-trading interval and that order-book evolution can contain information about opening prices. [source]

5. Opening Range / Initial Balance Skill Family

- ORB-5 / ORB-15 / ORB-30: define the first N-minute high/low and test breakouts with relative volume, VWAP, delta and breadth confirmation.
- Failed ORB: breakout beyond the range followed by acceptance back inside; test reversal only after confirmation.
- Initial Balance expansion: measure how far price extends from the early range and whether volume migrates with the move.
- Gap + ORB: separate gap-aligned breakouts from gap-opposed breakouts. They are different distributions and should not be pooled.
- ORB research is mixed by universe and implementation: academic work finds positive results for some mechanical constructions, while recent large-sample research can find near-breakeven performance. Hermes should therefore model ORB as a candidate feature, not an assumed edge. [source]

6. Volume Profile / Auction Market Skills

- POC: highest-volume price in the chosen profile.
- VAH/VAL: value-area boundaries; store the construction method because definitions vary by platform.
- HVN/LVN: high- and low-volume nodes used as structural context.
- Failed Auction: probe outside value, then return inside; combine with CVD/OFI and acceptance tests.
- Value Migration: track movement of POC/value area. Rising migration can support bullish acceptance; falling migration can support bearish acceptance.

7. Footprint, CVD and Absorption Skills

- Delta: Ask-traded volume minus Bid-traded volume.
- CVD: cumulative delta. Use price/CVD divergence as a conditional signal, not a guarantee.
- Absorption: high aggressive volume at an extreme with limited price extension. Confirm with repeated tests and replenishment.
- Stacked imbalance: useful as a feature, but fixed rules such as 300% or 'three levels' must be empirically calibrated.
- The source's descriptions of footprint imbalance and absorption are useful as market-structure hypotheses; they should be validated with local trade-level data rather than treated as proof of institutional activity. [source]

8. Microstructure: OFI, MLOFI, Microprice and Hawkes

Feature	Definition	Hermes use
OBI	$(BidVol - AskVol) / (BidVol + AskVol)$	Queue pressure
OFI	Event-based bid queue change - ask queue change	Short-horizon price impact
MLOFI	Weighted OFI across levels	Depth-aware pressure
Microprice	$(Ask * BidSize + Bid * AskSize) / (BidSize + AskSize)$	Fair-value tilt
Spread	Ask-Bid	Liquidity / execution cost
Cancellation rate	Cancels / quote events	Liquidity fragility

Feature	Definition	Hermes use
Replenishment	New size after consumption	Potential absorption
Trade intensity	Events or volume per unit time	Activity regime
Hawkes intensity	Self/cross-exciting order/event intensity	Event clustering

- Cont, Kukanov and Stoikov found that over short intervals, price changes in their NYSE sample were mainly driven by order-flow imbalance, with impact depending on market depth. The relationship is a research foundation, not a universal NSE coefficient. [source]
- Hawkes-process models capture self-exciting order arrivals and jump clustering. They are promising for event-intensity features, especially around volatility bursts and liquidity stress. [source]

9. Options Intelligence

- IV-RV: compare implied volatility with realized/forecast volatility.
- Surface: model $IV(K,T)$, skew, smile, term structure and changes in slope/curvature.
- GEX: estimate gamma-related hedging pressure, but explicitly model the assumptions used to infer participant positioning from public OI.
- Gamma Flip: estimated price where the modelled aggregate gamma changes sign.
- Vanna/Charm: secondary exposures that can help explain changes in hedge requirements as volatility and time change.
- PCR: useful as a context feature; research shows option ratios can carry information, but the relationship is horizon- and market-dependent. [source]

10. Statistical Arbitrage & Relative Value

- Cointegration: Engle-Granger and Johansen.
- Dynamic hedge ratio: Kalman filter.
- Spread dynamics: Ornstein-Uhlenbeck; estimate mean reversion speed and half-life.
- Basis: spot/futures and economically linked futures relationships.
- Calendar spreads: compare expiries after carry, liquidity and event adjustments.
- Never assume stationarity is permanent. Hermes should monitor rolling stability and disable the strategy when the relationship breaks.

11. Trend, Momentum and Volatility Breakout

- Donchian: N-period high/low breakout.
- Time-series momentum: $\text{sign}(P_t - P_{t-N})$ over several horizons.
- Volatility breakout: require range/ATR expansion beyond a regime threshold.
- Range compression: detect unusually compressed realized range, then test expansion.
- Relative strength: compare target return with benchmark/sector.
- Trend-following should be paired with a regime filter; a breakout strategy is often weakest when the market is range-bound.

12. Mean Reversion

- VWAP reversion: deviation / ATR.
- Z-score reversion: standardized distance from rolling equilibrium.
- Gap-to-close reversion: gap-fade candidate when opening acceptance fails.
- Post-exhaustion reversal: high flow with poor price continuation.
- Spread reversion: cointegrated spread Z-score.

13. Cross-Market Lead-Lag & Event Intelligence

- Track GIFT NIFTY, global equity futures, USDINR, crude, rates, sector indices and breadth.
- Use only information available before the prediction timestamp. The 2026 cross-index gap study explicitly time-aligns international markets before forecasting the target open, and finds opening-gap series with heavy tails and volatility clustering. [source]
- Event features: RBI, Fed, CPI, payrolls, Budget, major earnings, elections, index rebalances and other scheduled risk.
- Build an event-volatility gap: expected movement from IV/forecast compared with historical movement following similar events.

14. ML / Deep Learning Skill Stack

- Baseline: logistic regression, ridge/lasso, calibrated gradient boosting.
- Tree models: XGBoost/LightGBM.
- Sequence: LSTM/GRU/temporal CNN/Transformers.
- LOB: DeepLOB-style CNN + temporal model when true L2/trade data are available.
- Point process: Hawkes intensity as a feature or hybrid component.
- Meta-model: predict expected trade quality using component model outputs.
- Prefer probabilities such as $P(\text{return} > \text{threshold})$, Up/Flat/Down, movement magnitude, target-before-stop and realized volatility. Do not make exact next-price prediction the primary objective.

15. Validation and Anti-Overfitting Stack

- Walk-forward: rolling time-respecting train/validation/test.
- Purging: remove training observations whose labels overlap the test window.
- Embargo: leave a gap after test observations.
- CPCV: generate multiple out-of-sample paths when many candidate strategies are tested.
- Deflated Sharpe: adjust for selection bias, multiple trials and non-normal returns. It is a diagnostic, not a universal pass/fail threshold. [source]
- Stress: multiply spread/slippage, add latency, remove best trades, test regime changes and missing data.

16. Risk Engine & Position Sizing

- $\text{PositionSize} = \text{MaxRiskPerTrade} / (\text{StopDistance} * \text{ValuePerPoint})$.
- Start conservatively with fixed-fractional risk. Fractional Kelly can be tested only after probability calibration is strong.
- Use structural, ATR/volatility, time, probability and information stops.
- Hard controls outside ML: maximum daily loss, trade loss, notional, concentration, gamma/vega, slippage and drawdown.
- Do not import the historical source's aggressive 9-10% per-trade risk into Hermes. The supplied tests themselves show that gap strategies can suffer very large drawdowns. [source] [source]

17. Execution & Market Making

- TWAP: time-sliced execution.
- VWAP: volume-aware execution.
- POV: target participation rate.
- Iceberg: partial display where the venue/broker supports it.
- Almgren-Chriss: balance impact against execution-risk cost.

- Avellaneda-Stoikov: advanced inventory-sensitive market making; keep it separate from Hermes V1 directional trading.
- Institutional FPGA/co-location microsecond architecture is useful as a reference for HFT design, but not a requirement for a compliant retail/API Hermes implementation.

18. Hermes Gap Classifier - Practical Algorithm

1. Detect gap at NSE open. 2. Compute GapPct and GapATR. 3. Read pre-open imbalance and its persistence. 4. Check scheduled news/event risk. 5. Measure first 5-15 minute acceptance or rejection. 6. Compute OFI, microprice tilt, CVD and VWAP position. 7. Check breadth, sector and futures confirmation. 8. Classify: - Acceptance + confirming flow -> GAP-AND-GO candidate. - Rejection + return toward prior close -> GAP-FILL candidate. - Conflicting evidence -> WAIT. 9. Calculate expected value after spread, slippage and fees. 10. Risk engine sizes or vetoes the trade. 11. Log the full feature snapshot and model version.

19. Hermes Skill Registry

Tier	Skill	Purpose
P0	Gap-fill vs gap-go classifier	Convert the uploaded gap research into a two-regime decision problem.
P0	NSE pre-open imbalance	Exploit current auction information without copying obsolete US thresholds.
P0	ORB / Initial Balance	Simple, explainable opening regime baseline.
P0	OFI + microprice	Short-horizon order-flow intelligence.
P0	VWAP + relative volume + breadth	Low-complexity confirmation layer.
P1	Volume profile / failed auction	Structural context.
P1	CVD / absorption	Flow confirmation.
P1	IV-RV / volatility surface	Option relative value.
P1	GEX / gamma flip estimate	Volatility/momentum regime context.
P1	Cross-market lead-lag	Pre-open and intraday external information.
P2	HMM/GARCH	Probabilistic regime state.
P2	Kalman / cointegration	Relative-value book.
P2	Hawkes intensity	Flow/jump clustering.
P2	DeepLOB	Only after high-quality L2 data and strong baselines.
P3	Market making	Separate strategy/business model.

20. Hermes Training Protocol

- Skill Card: name -> hypothesis -> data -> features -> trigger -> confirmation -> exit -> risk -> execution -> failure modes -> validation -> deployment status.
- Train each skill independently, then test incremental value when combined with the regime engine.
- Maintain an experiment ledger recording every trial; the number of attempts matters for interpreting final performance.
- Promote a skill only after it survives multiple time windows, realistic costs, parameter sensitivity and an untouched holdout.
- Make NO-TRADE a first-class action.

21. What Counts as a 'Cheat' in Hermes

- Richer information: pre-open auction state, order flow, option surface and cross-market data.
- Better state inference: regime, event and liquidity conditions.
- Better short-horizon modelling: OFI, microprice, queue dynamics and Hawkes intensity.
- Better option modelling: IV-RV, skew, gamma/vanna/charm and scenario Greeks.
- Better validation: purged/embargoed CV, CPCV and DSR-aware research accounting.

- Better execution: slippage-aware order placement.
- Better risk: independent controls and the ability to refuse a trade.

22. Final Operating Rule

Information -> Features -> Regime -> Probability -> Expected Value -> Risk -> Execution -> Audit.

The new gap source is most valuable when treated as a library of hypotheses: it adds gap reversal, gap continuation, gap age, pre-open volume and gap-fill dynamics. The modern Hermes layer should then test those hypotheses against current NSE market structure and combine them with auction, microstructure, volatility, options and regime information.

Verified / Research Sources

Supplied PDF: Active Trader gap strategy collection (historical US material): Morning Reversal, Overnight Gap, Opening Gap, Gap Closer. The source itself warns that its gap systems are experimental and that past performance may not reflect real-time behaviour. [source]

NSE - Market Timings: <https://www.nseindia.com/static/market-data/market-timings>

NSE - Equity Derivatives Pre-Open Session: <https://www.nseindia.com/static/products-services/equity-derivatives-pre-open-session>

NSE - Current Equity Derivatives Contract Specifications:
<https://www.nseindia.com/static/products-services/equity-derivatives-contract-specifications>

SEBI - FY25-FY26 individual equity-derivatives profitability study, Aug 20, 2026: <https://www.sebi.gov.in/>

Cont, Kukanov & Stoikov - The Price Impact of Order Book Events: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=1712822

Zhang, Zohren & Roberts - DeepLOB: <https://arxiv.org/abs/1808.03668>

Moskowitz, Ooi & Pedersen - Time Series Momentum: <https://doi.org/10.1016/j.jfineco.2011.11.003>

Heston - A Closed-Form Solution for Options with Stochastic Volatility: <https://doi.org/10.1093/rfs/6.2.327>

Almgren & Chriss - Optimal Execution of Portfolio Transactions: <https://doi.org/10.21314/JOR.2001.041>

Bailey & Lopez de Prado - The Deflated Sharpe Ratio: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2460551

Soebhag - Option Gamma and Stock Returns: <https://doi.org/10.1016/j.jempfin.2023.101442>

Andersen & Bondarenko - VPIN and the Flash Crash / VPIN dispute: <https://doi.org/10.1016/j.finmar.2013.05.005>

Madhavan & Panchapagesan - Price Discovery in Auction Markets: <https://doi.org/10.1093/rfs/13.3.627>

Opening Range Breakout research: <https://doi.org/10.1016/j.frl.2012.09.001>

2026 cross-index gap research: <https://doi.org/10.1016/j.bir.2026.100871>

Hawkes process order-flow / price-impact research: <https://doi.org/10.1017/S1446181119000038>

Evidence grading

Verified current: exchange/regulator facts and current session mechanics. Verified research phenomenon: relationship shown in a defined dataset or paper. Candidate skill: plausible and testable feature/heuristic. Not a rule: historical thresholds, benchmark scores, or causal statements that do not establish universal profitability.